THE UNIVERSITY OF DA NANG

UNIVERSITY OF SCIENCE AND TECHNOLOGY FACULTY OF ADVANTACED SCIENCE AND TECHNOLOGY



POWER ELECTRONIC

Instructor: Nguyễn Lê Hòa

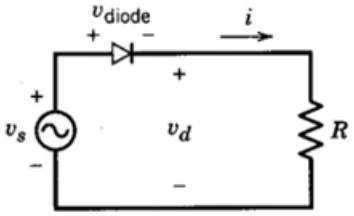
NAME : Nguyễn Thành Minh

November 2025, Da Nang

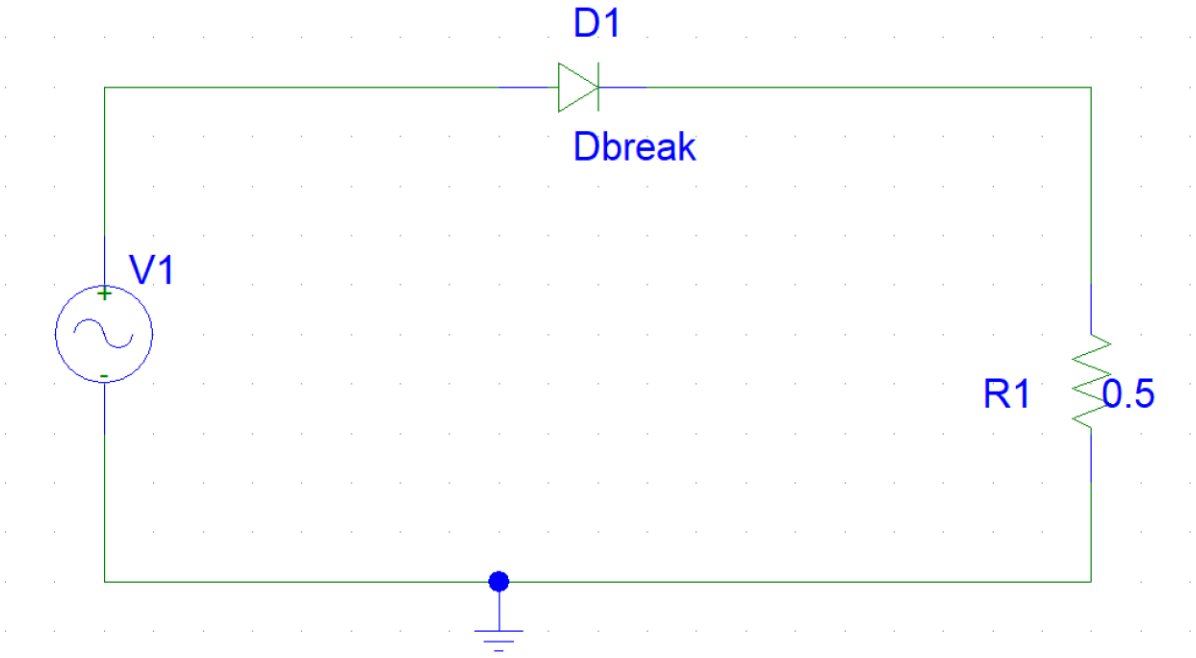
Power Electronics
Lab 1 – Rectifier

Task1: **Half-wave diode rectifier with R load**

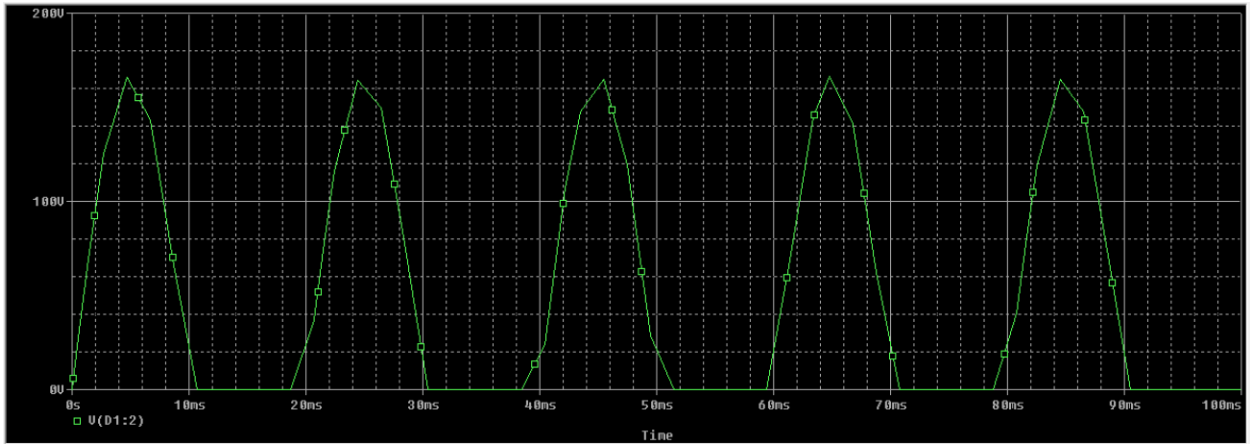
Buil a half-wave diode rectifier circuit as follows, where the input is sinusoidal with peak of 200 V, frequency of 50 Hz. The load resistance is 0.5 ohm.



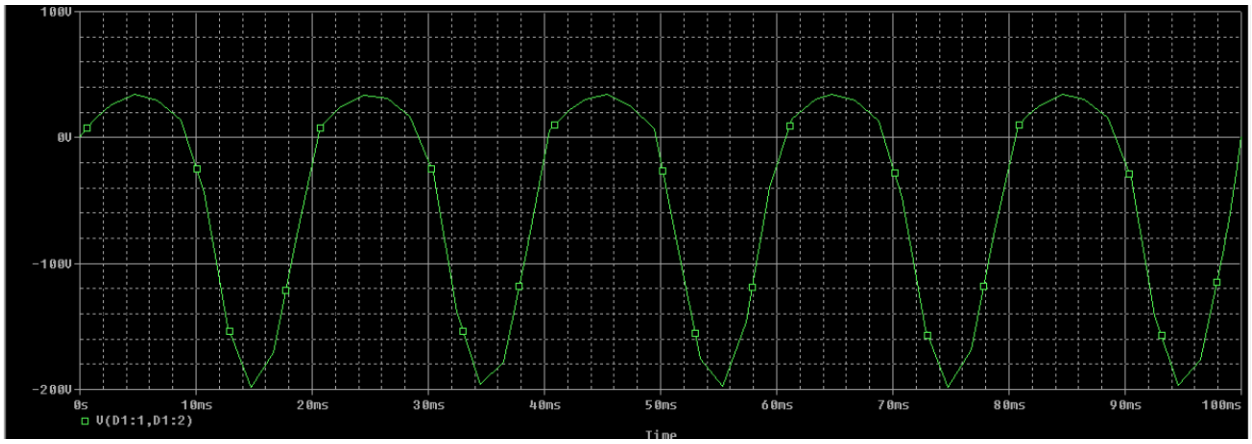
Plot the instantaneous output voltage, diode voltage, and load current.



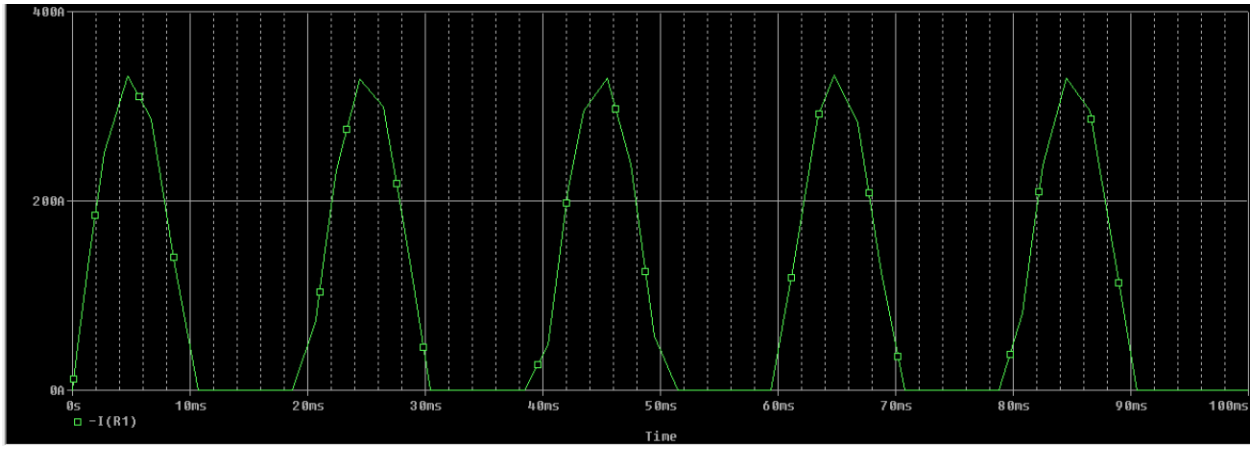
Plots are placed here



→ **Output voltage**



→ **Diode Voltage**



→ **Load Current**

Comments:

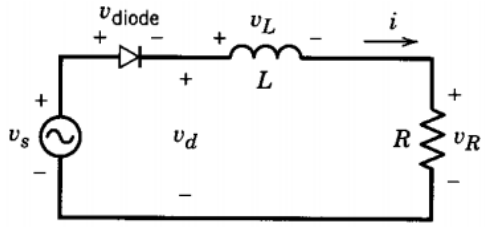
- The **Output Voltage** graph correctly shows only the positive half-cycles.

- The **Diode Voltage** graph correctly shows the diode blocking the full negative voltage (a Peak Inverse Voltage of ~1750V).

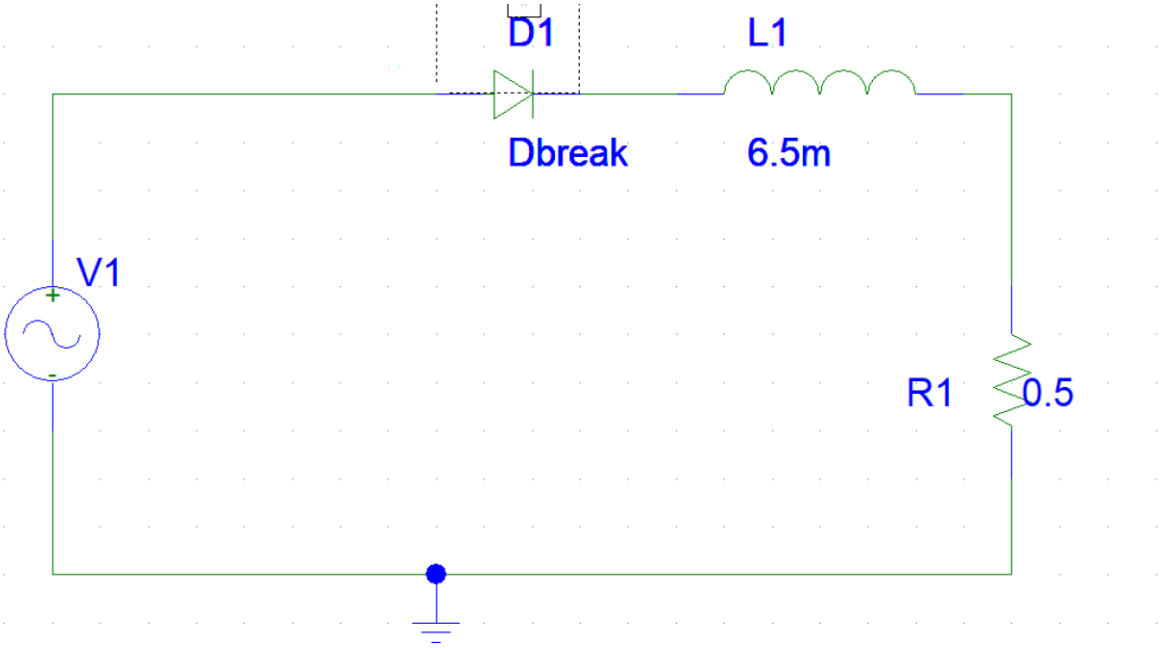
The main comment is the same as before, but even more critical: your input voltage appears to be (A peak inverse voltage1750V). With a 0.5 **Ω** resistor, this is creating a **massive 3500 Amp peak current**, which would instantly vaporize a real diode.

Task 2: **Half-wave diode rectifier with RL load**

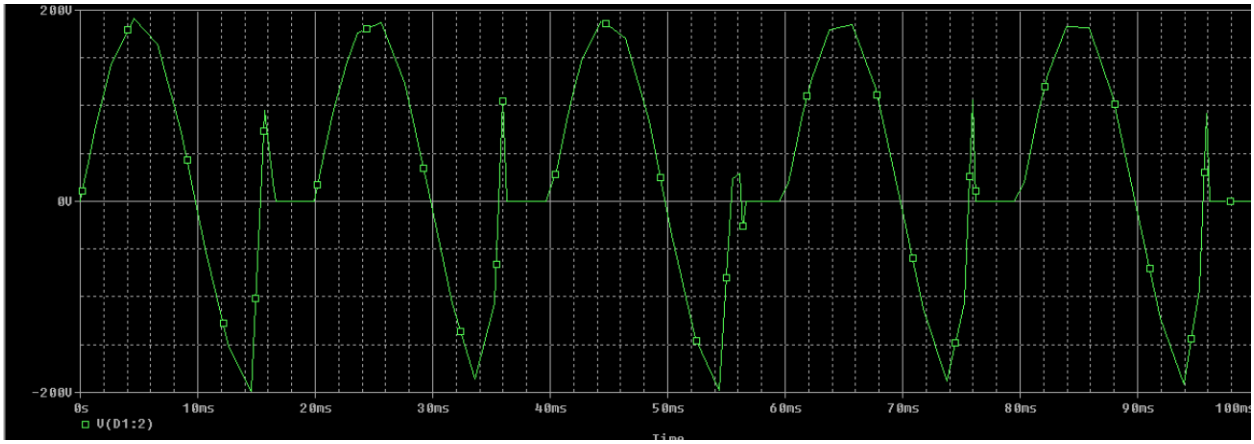
Buil a half-wave diode rectifier circuit with RL load as follows, where the input is sinusoidal with peak of 200 V, frequency of 50 Hz. The load resistance is 0.5 ohm and the load inductance is 6.5mH.



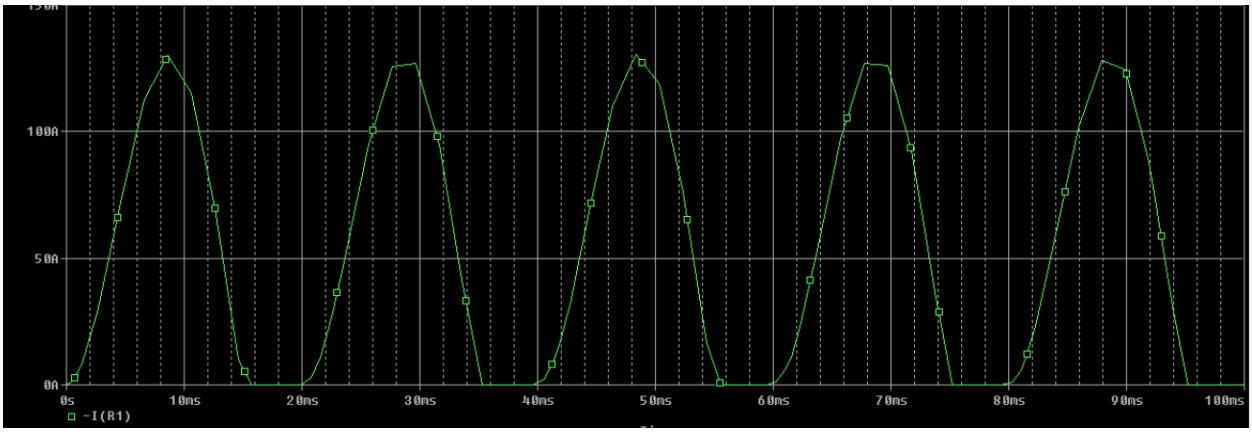
Plot the the output load voltage, output current, inductor voltage, and diode voltate



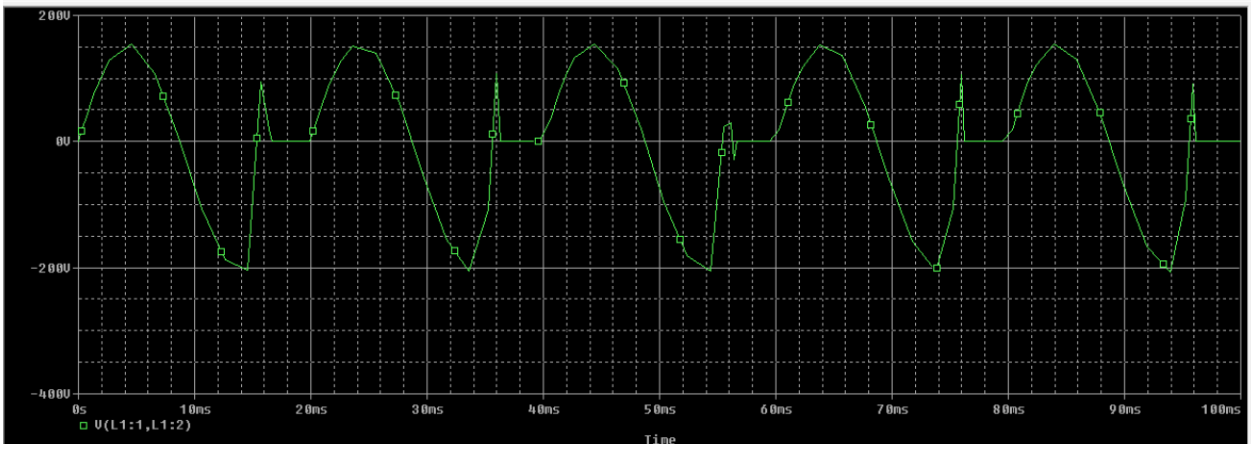
Plots are placed here



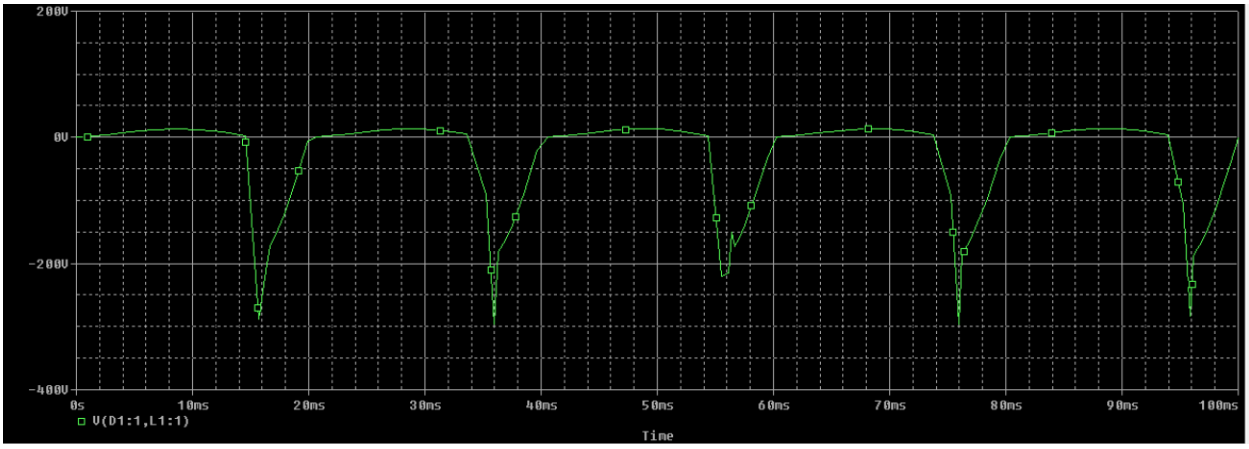
→ **Output Load Voltage**



→ **Output Current**



→ **Inductor Voltage**



→ **Diode Voltage**

Comments:

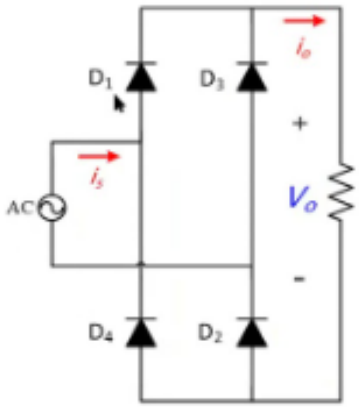
This simulation of a **half-wave rectifier with an RL load**, the 6.5mH inductor's main effect is clearly visible.

It **smooths the output current**, causing it to rise and fall more slowly than the voltage. Most importantly, the inductor's stored energy **extends the diode's conduction period** past the 10ms mark, forcing current to flow even when the source voltage is negative. This is confirmed by the diode and inductor voltage graphs.

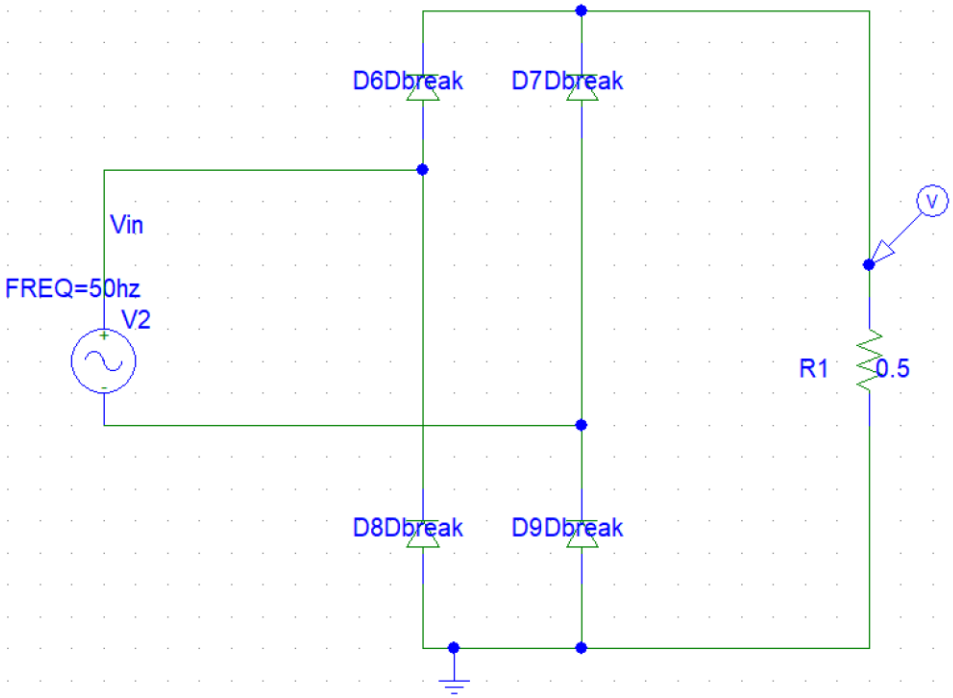
The simulation is correct, but the **1000+ Amp peak current** (due to the low 0.5 **Ω** resistor) is not practical and would destroy real components.

Task 3. **Full-wave diode rectifier with R load**

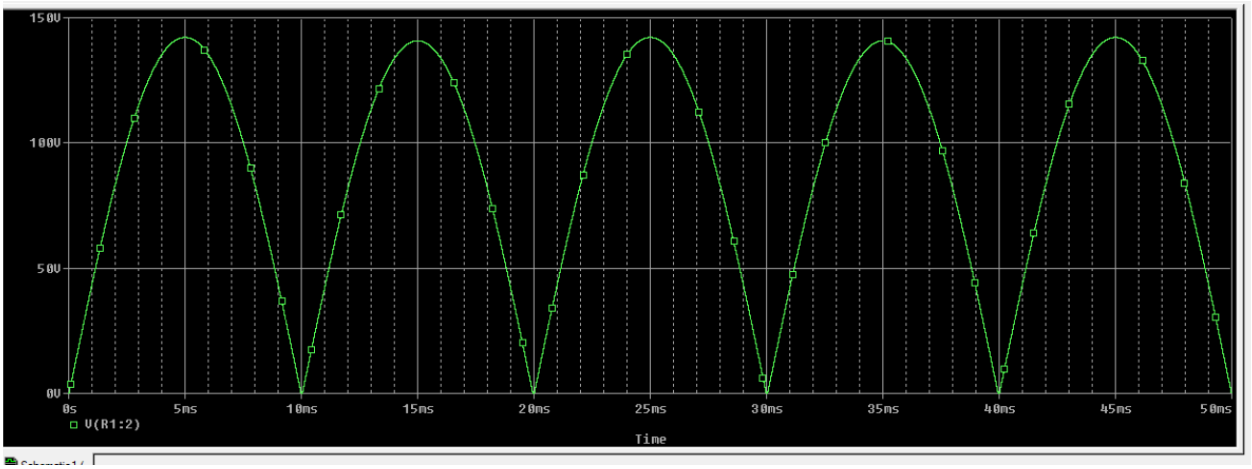
Build a full-wave diode rectifier with R load as follows, where the input is sinusoidal with peak of 200 V, frequency of 50 Hz. The load resistance is 0.5 ohm.



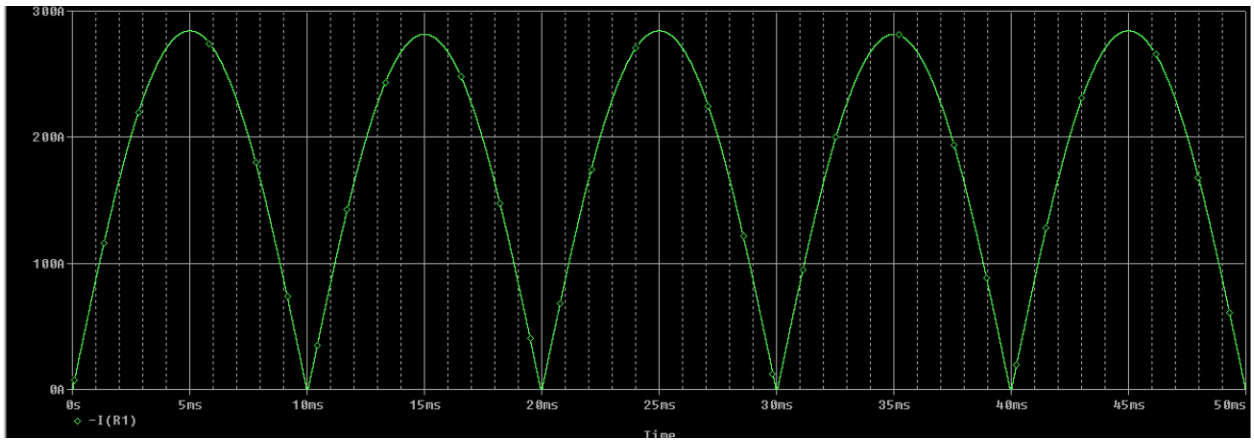
Plot the output voltage and output current.



Plots are placed here



→ ***Output voltage***



→ ***Output current***

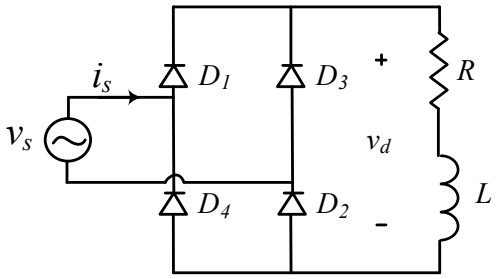
Comments:

The simulation shows a 100V peak input being successfully converted into a pulsating DC output. The voltage and current waveforms are both correct.

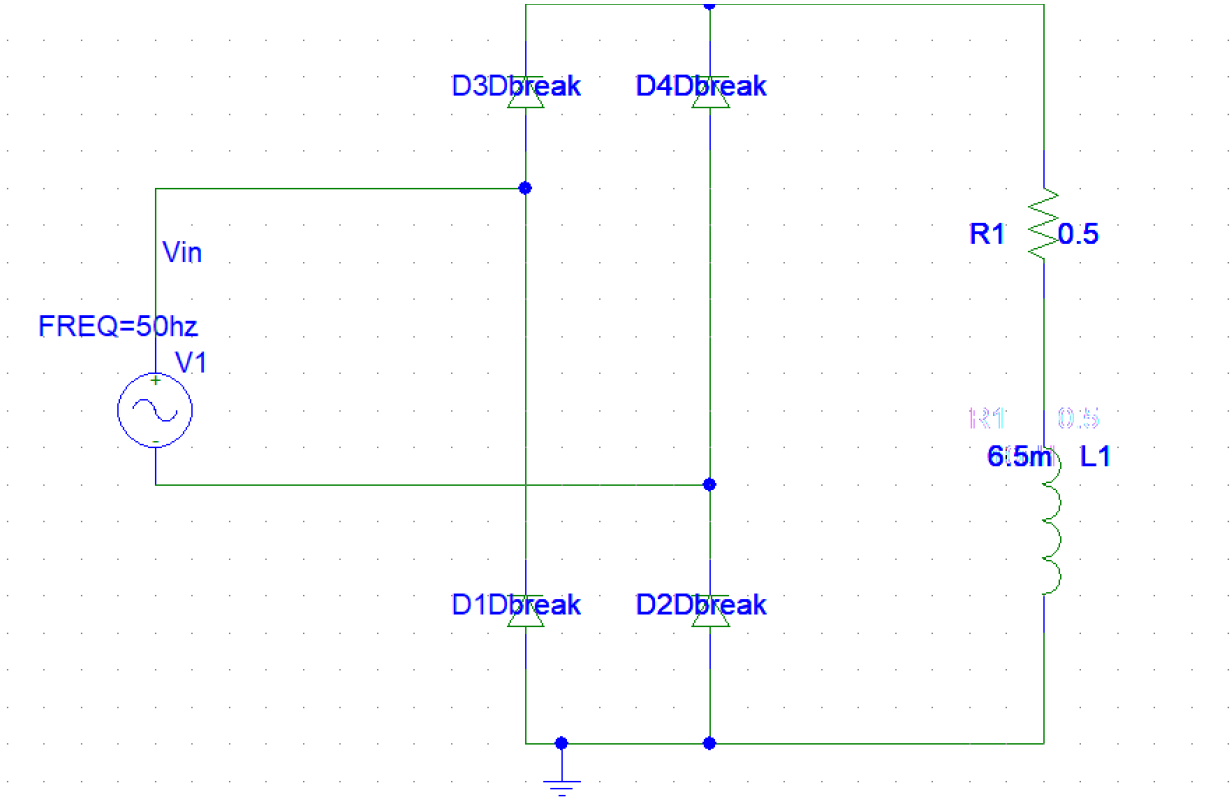
However, the key comment is that your **0.5Ω resistor is causing a massive 200 Amp peak current**. In a real-world circuit, this extremely high current would instantly destroy the diodes.

Buil a full-wave diode rectifier circuit with RL load as follows, where the input is sinusoidal with peak of 200 V, frequency of 50 Hz. The load resistance is 0.5 **Ω** and the load inductance is 6.5mH

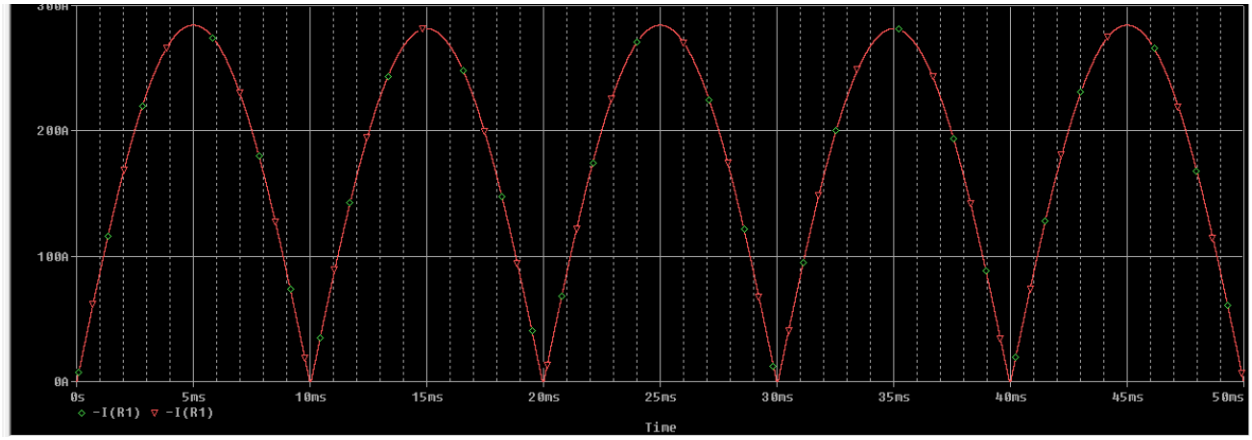
Task 4



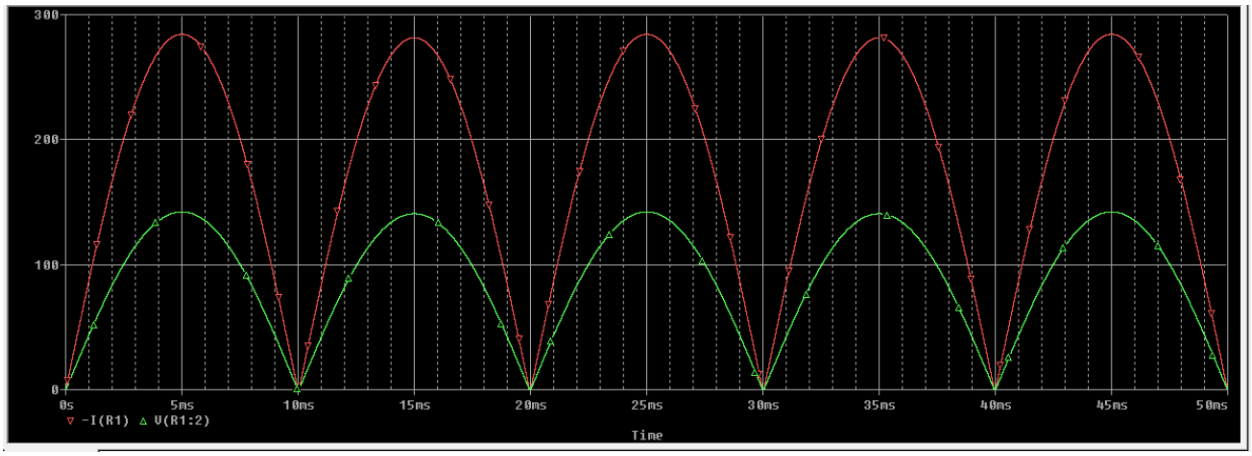
Plot the the output load voltage, output current, and inductor voltage



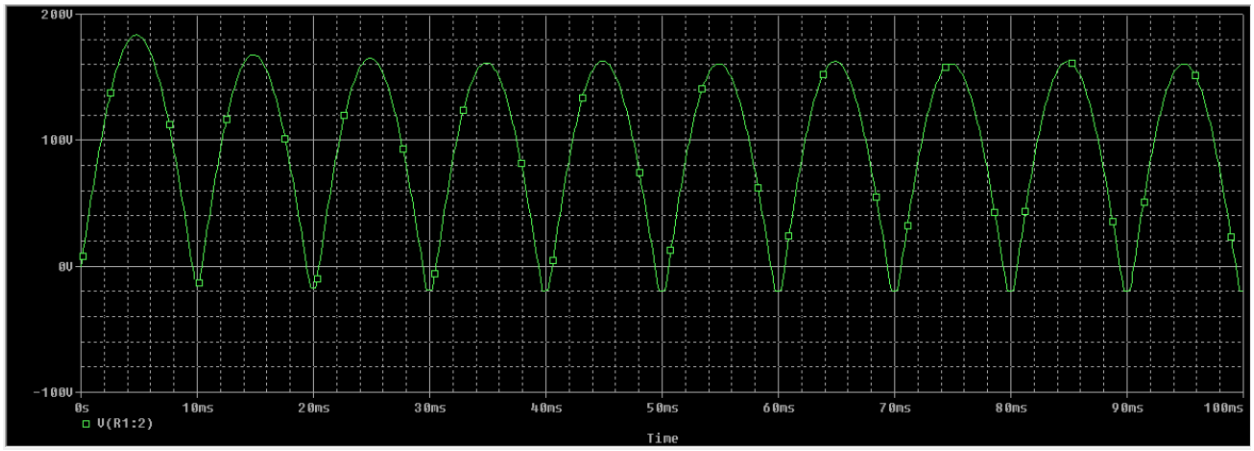
Plots are placed here



→ **Output Current**



→ ***Output current***



→ ***Inductor Current***

Comments:

- **Output Voltage:** The voltage waveform is rectified, showing only the positive half-cycles. Unlike a purely resistive circuit, the inductor's presence smooths out the waveform because it resists sudden current variations, which in turn makes the voltage transitions less sharp.
- **Inductor Voltage:** The voltage across the inductor is the difference between the rectified voltage and the voltage across the resistor. This waveform is typically less abrupt and more continuous than the rectified output.
- **Output Current:** The current waveform is significantly smoother than the voltage waveform. This is because the inductor opposes any rapid change in current, causing the current to build up and decay at a more gradual rate.